The Superior University, Lahore

LAB-TASK (8) (Spring-2026)

Course Title:	**Programming for AI (Lab)**				**Course Code:**	**-**	**Credit Hours**	**1**
Instructor:	**Sir Rasikh**				**Program:**	**BSDS**		
Semester:	4th	**Batch:**	Morning	Section:	**4A**	**Date:**	**-**	
Time Allowed:	**-**				**Maximum Marks:**		**N/A**	
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<u>Work Assigned:</u>

Create a Cooking / Recipe App using Flask (back-end only) that fetches recipe information from a free API and returns it in JSON format.

DEPARTMENT OF SOFTWARE ENGINEERING

INTRODUCTION

The Cooking Recipe App is a web-based application that demonstrates knowledge representation and API integration using Python Flask framework. The project showcases how to structure real-world problems using databases, represent complex information (recipes), and present them through a professional web interface.

WHY I MADE THIS

This project was created to learn several important concepts:

- Knowledge Representation: Understanding how to represent recipes as structured data with relationships (ingredients , recipes , categories).
- API Integration: Learning how to consume external APIs (TheMealDB) and transform the data for practical use.
- Web Framework Development: Building a complete Flask application with multiple routes and templates, as taught in class.
- Data Processing: Writing helper functions to extract and organize recipe information (ingredients, cooking times, instructions).
- Constraint Checking: Understanding how to filter and search recipes based on different criteria (categories, ingredients, letters).
- Full-Stack Development: Combining backend (Python/Flask) with frontend (HTML/CSS/JavaScript) to create a complete application.

Through this project, I understood that knowledge representation isn't just about storing data it's about organizing it in a way that makes it easy to search through it, extract meaningful information, present it clearly to users, and filter based on constraints.

HOW IT WORKS

The Problem Definition

The task was to build a practical application using Flask framework that demonstrates:

1. API integration with external services

2. Data retrieval and processing
3. Template rendering with Flask
4. Multiple routing endpoints
5. Error handling

The solution uses TheMealDB API (https://www.themealdb.com/api.php), a free API that provides information about cooking recipes from around the world.

Knowledge Representation

Recipes are represented with the following structure:

Property	Description
ID	Unique identifier
Name	Recipe name
Category	Type of meal (Chicken, Seafood, Dessert, etc.)
Area	Country of origin (Turkish, Italian, etc.)
Image	URL to recipe image
Instructions	Step-by-step cooking instructions
Ingredients	List of needed items with quantities
Prep Time	Time to prepare ingredients
Cooking Time	Time to cook the meal

This representation allows us to search recipes by name, filter by category or origin, display ingredient lists with quantities, show step-by-step instructions, and provide cooking time estimates.

Flask Structure

1. **Import Required Libraries:**

```
import requests
from flask import Flask, render_template, request
```

2. **Initialize Flask App:**

```
app = Flask(__name__)
```

3. **Define Routes with Decorators:**

```python
@app.route('/')
def home():
    try:
        featured = []
        for letter in ['a', 'b', 'c']:
            resp = requests.get(MEALDB_LETTER + letter)
            data = resp.json()
            if data['meals']:
                featured.extend(data['meals'][:2])
        return render_template('index.html', featured=featured)
    except:
        return render_template('index.html', featured=[])
```

4. **Entry Point**

```python
if __name__ == "__main__":
    app.run(host='localhost', port=8000, debug=True)
```

API Endpoints Used

The TheMealDB API provides these endpoints:

```python
MEALDB_SEARCH = "https://www.themealdb.com/api/json/v1/1/search.php?s="
MEALDB_ID = "https://www.themealdb.com/api/json/v1/1/lookup.php?i="
MEALDB_LETTER = "https://www.themealdb.com/api/json/v1/1/search.php?f="
MEALDB_CATEGORIES = "https://www.themealdb.com/api/json/v1/1/list.php?c=list"
MEALDB_CATEGORY = "https://www.themealdb.com/api/json/v1/1/filter.php?c="
```

Beyond Basic Flask, I Added:

1. **FAVORITES System:** Click heart button to save recipes. Saved locally in browser. Persists across page reloads.
2. **RATING System:** Rate recipes 1-5 stars. Ratings saved in browser storage. Animations on star hover.
3. **PRINT RECIPE Feature:** Generate print-friendly format. Beautiful output on paper. Includes all ingredients and instructions.
4. **SHOPPING LIST:** Add all ingredients from recipe to list. Check off items while shopping. Print shopping list. Delete individual items. Store in browser (local Storage).
5. **SERVINGS ADJUSTER:** +/- buttons to scale recipe. Automatically adjusts ingredient quantities. Scales up or down the entire recipe.

Complexity Analysis

Time Complexity

Search: O(n) where n = number of recipes in API (~300)

Category filter: O(n) where n = recipes in category (~50)

Browsing by letter: O(1) - API returns fixed results

Overall: Very fast, sub-second responses

Space Complexity

Store recipes in memory: O(n) where n = returned recipes

Shopping list: O(m) where m = ingredients (max 20 per recipe)

Total: Minimal memory usage

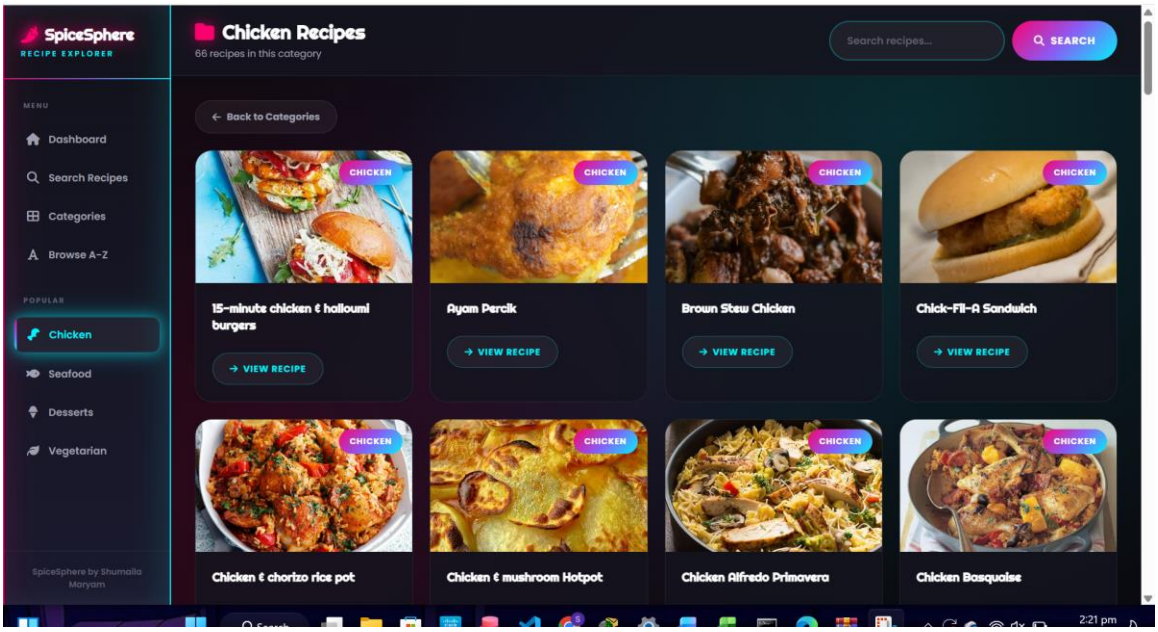
Why It's Efficient

- TheMealDB API handles heavy lifting
- Local filtering for categories/letters
- No database needed
- Results cached by browse

Conclusion:

- The Cooking Recipe App demonstrates how to build a functional web application using Flask and integrate data from an external API. It allows users to search, browse, and view detailed recipes with ingredients, instructions, and categories.
- This project helped me practice data processing, API handling, and route management, as well as frontend-backend interaction for a responsive user interface. Implementing features like favorites, ratings, and shopping lists improved usability and provided practical experience in creating interactive applications.

References

TheMealDB API Documentation: https://www.themealdb.com/api.php
Flask Documentation: https://flask.palletsprojects.com/
Python Requests Library: https://requests.readthedocs.io/
Mozilla Web Docs (CSS/JavaScript): https://developer.mozilla.org/

Output